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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Second Semester 2021-22 (Part-A, CLOSED BOOK)

EEE/CS/INSTR F241 Microprocessor programming & Interfacing

May 20, 2022

COMPREHENSIVE EXAMINATION - PART A

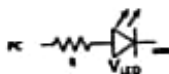
45 M, ~ 1.5 hour

- There are two parts in this exam. PART A is in-built; please write your answers in the blank/space provided. Ensure you have both sides of PART A printed.
- Once you turn in PART A after ~1.5 hour, PART B would be given.
- TAKE ALL DON'T CARE VALUES AS 0.
- Write all register values and memory locations/addresses in hexadecimal only.

Q1. The base address of 8255 is B7h. You wish to switch off a LED directly connected to PC5 of 8255 using BSR mode. Fill in the blanks which can be done to do the same [5M]

② Mov al, 0Ah [2M] (*take all don't care's as 0, mention values in hex only)

Out port, al [2M]



Which pins coming from 8086 will be connected to D0-D7 of 8255 (answer in format e.g. D4-D6)

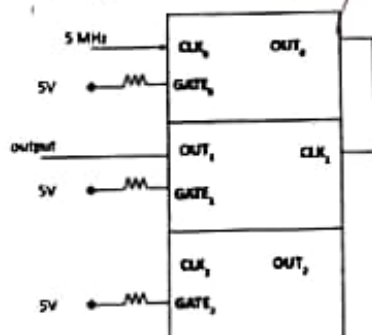
D0-D7 [1M]

Q2. An 8254 timer IC interfaced with 8086 is initialized as given in the code snippet below. Mention the frequency of the OUT0, OUT1 in kHz and the duty cycle (High duration/Total duration) in percentage for the output waveform. The starting address of timer is 80h. [8 Marks]

OUT0 Frequency = $\frac{2500}{2} = 1250$ kHz [2M] (mention in kHz)

OUT1 Frequency = 1 kHz [3M] (mention in kHz)

Duty Cycle = 50% [3M] (mention in percentage, e.g. 20%)



```
MOV al, 00110110b
MOV 86h, AL
MOV AL, 01110100b
MOV 86h, AL
MOV AL, 0E2h
MOV 80h, AL
MOV AL, 04h
MOV 80h, AL
MOV AL, 04h
MOV 82h, AL
MOV AL, 00h
MOV 82h, AL
```

Q3. You have a sensor whose output changes by 20mV for every 1 degree Celsius.

You need to select a suitable ADC for the application which requires to measure temperature at a resolution of 1 degree Celsius. You have an option to choose between ADC0808 and AD570. AD570 is connected with $V_{CC} = (+5V)$ and $V_{EE} = (-15V)$ and ADC 0808 is connected to a V_{ref} of 5V. 1Mhz clock input is given to ADC 0808. [5M]

(a) Which ADC will you prefer (AD570 or ADC 0808) for this application? Provide your answer with valid justification. [2.5M]

We will prefer ADC 0808 since its resolution = $\frac{5V}{256} = 19.6mV$ which is approximately equal to the given 20mV in the question.

(b) What is the conversion time for the ADC required for the application mentioned in part a. in microseconds? 4.5 [2.5M]

Q4. For 80386 processor working in protected memory mode of operation with paging enabled, the GDTR, LDTR, DS and CR3 value are given below:
 GDTR - 100020h LDTR - 0058h DS: 003Eh CR3: FF 00 00 00h [15M]
 If the instruction MOV EAX, [00 00 00 44h] is executed by the application, answer the following questions based on this information: (Please refer to the GDT, LDT, PD and PT tables in cheat sheet)

- Which of the table will be accessed to evaluate the final descriptor (GDT1/GDT2/LDT1/LDT2)? LDT1 [1M]
- Which entry number in the suitable table will be accessed to find the final descriptor for evaluating the linear address. 11th [1M] (Consider the entry at base address as 0th number entry)
- What are the privilege levels of the segment which this application can access? 10 4 11 [1M]
- Which address of GDT table will be accessed in course of evaluating the descriptor? 100078 [2M]
- What is the descriptor value for evaluating the linear address? 0 1 81 00 00 [2M]
- What will be the linear address? 01 81 00 44 [2M]
- What will be the physical address accessed in the memory? 15 00 00 44 [5M]
- What are the privilege levels of the application which can access this segment? 00-1 [1M]

Q5. Suppose that you are interfacing 3 interrupt sources to 8086. One interrupt source is connected to the NMI pin and other two are connected to IR6 and IR7 pin of 8259. IR7 should be assigned to priority level 0 and IR6 should be assigned to priority level 7. Specific EOI should be given for IR6 and IR7. IR6 is assigned type number 46h. Base address of 8259 is 60h. All interrupt sources raise edge triggered interrupts. [12 Marks] (*Note: PUSHA/ POPA is used to Push/ Pop together AX, CX, DX, BX, original SP, BP, SI, and DI)

Write the code snippet for initialization/configuration of 8259 (i.e. the required ICWs and OCWs) as per the requirement and the ISR for the NMI and the ISR for the NMI and IR7.

- The NMI should set the bit number zero of the value in memory location 80h without affecting other bits (assume that 80h is I/O mapped memory block which can be accessed like an I/O device).
- The IR7 should set the bit number 1 of the value in memory location 80h without affecting other bits (assume that 80h is I/O mapped memory block which can be accessed like an I/O device).
- The ISR should be written so that nested interrupt handling is enabled.
- All IR's except IR6 and IR7 should be masked.
- All the register contents before and after the interrupt execution should be the same

Write values of ICWs/ OCWs/ address etc. in hex format only. DON'T CARES/ IRRELEVANT BITS SHOULD BE TAKEN AS 0 STRICTLY.

ICW & OCW initialization [5M]	ISR: NMI [4M]	ISR: IR7 [3M]
ICW1:- mov dx, 60h mov al, 13h out dx, al ICW2:- mov dx, 62h mov al, 80h out dx, al ICW4:- mov dx, 62h mov al, 11h out dx, al OCW1:- mov dx, 62h mov al, 8Fh out dx, al OCW2:- mov dx, 60h mov al, 66h out dx, al OCW3:- mov dx, 60h mov al, 06h out dx, al	mov dx, 80h mov al, 00h out dx, al	mov dx, 80h mov al, 01h out dx, al

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COMPREHENSIVE EXAMINATION - PART B

35 M, ~ 1.5 hour

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- Read all questions carefully
 - **TAKE ALL DON'T CARE VALUES AS 0.**
 - Write all register values and memory locations/addresses in hexadecimal only.
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Q1. Design the memory interfacing for an 80286 based system which has **1024 KB** of memory of which **256 KB** is **ROM** and the rest is **RAM**. The starting address of RAM is **F0 00 00H**. ROM begins from the next address after the last address of RAM. Only the following hardware components are available for this interfacing:

27256 chips- 8 Nos.

61512 chips- 12 Nos.

LS138- 1 No.

Logical gates:

2-input OR gates -24 Nos.

4-input AND gate – 1 No.

Inverters – 2 No.

The system is expandable in nature. Assume MEMR' and MEMW' signals are available already.

- a) Show the address mapping for all the memory chips clearly in the form of a table in the space provided in the built-in answer sheet. **[5M]**
- b) Draw a neat memory interfacing logic showing all the required signals. If the system uses LS138, show all the inputs and outputs of the LS138. Further if an LS138 is used, just show the interfacing of one even and odd bank each of ROM and RAM with the LS138 and 80286. **[9M]**

(PTO for question 2)

Q2. Design an 8086-based **Vehicle detection system (VDS)** for a one-way military road in an army cantonment zone. In this scenario, military vehicles moving on the road wirelessly broadcast their vehicle identity (id) and speed data as two 4-bit values. The VDS is deployed at one side of the road in a particular location on the road. The wireless data receivers connected to 8086 through 8255 receive the vehicle identity (id) and the speed data (from the vehicles passing through) wirelessly and pass on these data values to 8086 in parallel. As soon as a vehicle is detected, an edge-triggered interrupt **INT_DETECT** is raised to 8086 by the wireless receiver. Note that the system should also keep a count of the number of vehicles which have passed the VDS in a day using 8254. Assume the maximum count of vehicles will not exceed 60000 at any point of time. Assume that the counter is manually reset every day. On vehicle detection, the vehicle identity and speed data need to be read. Assume only a single vehicle is moving on the road in a single direction at a particular time.

Also assume there is a comparator connected to the wireless data receiver. In case, the speed limit exceeds a certain threshold as detected by the comparator, an interrupt **INT_LIMIT** is raised to 8086. Assume that **INT_LIMIT** has a higher priority over **INT_DETECT**. As soon as **INT_LIMIT** is raised, an alarm system needs to be activated. To activate the system, a square wave signal of 1KHz needs to be sent to the alarm system using 8254. Note that this alarm should be active for 10 seconds. **Only port C of 8255 is available for the alarm interface.** *1 kHz square wave is the only signal provided to the alarm system. Only Port B should be used for wireless data receiver.*

An 8259A chip is available for servicing interrupts in the system. Assume it is configured in edge triggered mode, non-buffered in special fully nested mode. Assume default priorities for the interrupts with IR0 having the highest priority and IR7 having the lowest priority. The interrupt sources in this system are connected from IR0 onwards and all unused interrupt source pins are masked. Further, 8259A is configured in AE01 mode. E.g., If IR5 is used and an interrupt is generated on it, an interrupt of type 65H is raised to 8086.

Components available: 8086 - 1 Nos., 8255 - 1 Nos., 8254 - 1 Nos, 8259A - 1 Nos, LS138 - 1 No. **Assume that port A of 8255 is reserved for some other task where it is taking input and not available for this particular system design. Also, the NMI pin of 8086 is reserved for some other purpose and cannot be used here.** The base addresses of 8254, 8255 and 8259A are FFE5H, FF6DH and FF1DH respectively. Assume an 8284 is available for providing a clock frequency of 5MHz to 8254. **Please take any Don't Care values (X) as 0s while solving the question. Mention values of registers, various values etc. in HEX in the code/ ISRs.** **[21M]**

The solution for this design problem needs to be in the following format:

- Draw the hardware interfacing for all the I/O peripherals (both the system side (8086) and I/O side) discussed above. **[6M]**
- Write the code snippets for the initialization of each of the I/O peripherals:
 - 8255, ii) 8254, iii) 8259A **[2M+6M+3M]**
- Write the Interrupt Service Routines (ISRs) for all the scenarios discussed above. **[1M+3M]**